

Philadelphia Urban Safety - Temporal & Spatial Crime Analysis

Project Summary: What the analysis is based on and how the final intelligence report is produced

1. Analysis Objective

The project is designed to identify **where and when selected property-related crimes occur most frequently** within approximately 500 meters of central Philadelphia coordinates (39.9526, -75.1652). It combines spatial filtering with time-based analysis to highlight concentration, peak hours, peak days, and the most common incident category.

2. Data Source and Scope

Component	Basis used in the code
Data source	Philadelphia CARTO SQL API (phl.carto.com)
Primary dataset	incidents_part1_part2; fallback table: inc_part1_part2
Time window	Previous 180 days (~6 months) from the date the notebook is run
Center point	Latitude 39.9526, Longitude -75.1652
Spatial scope	0.0045 degrees around the center point, used as an approximation of ~500 meters
Incident types	Theft from Vehicle, Other Theft, Burglary Residential, Burglary Commercial

3. How the Data Is Selected

The SQL query keeps only incidents that satisfy **all three conditions**: (1) the incident occurred during the last 180 days, (2) the incident belongs to one of the four selected theft/burglary categories, and (3) its geographic location falls inside the defined radius around the selected center point. Results are ordered chronologically by dispatch date and time.

4. Temporal Features Created

Feature	Purpose
Hour	Extracts the hour (0-23) to identify the time of day with the highest incident frequency.
Day of week	Converts each incident date into Monday-Sunday to compare patterns across weekdays/weekends.
Date	Keeps a date-only field for day-level analysis or future trend aggregation.

5. Visual Analysis Produced

Output	What it shows
Interactive 500m map	Plots every selected incident around the center point and draws the analysis boundary. Each marker identifies the incident type and hour.
Crime by hour chart	Counts incidents for each hour of the day to identify the peak crime time.
Day x Hour heatmap	Cross-tabulates day of week and hour to reveal combinations with the greatest concentration of incidents.

6. Final Intelligence Report - Calculation Basis

Total Incidents: Number of records returned after all date, crime-type, and spatial filters are applied.

Most Dangerous Day: The day of the week that appears most frequently in the filtered incident data (statistical mode).

Peak Crime Time: The hour of the day that appears most frequently in the filtered incident data (statistical mode).

Primary Pattern: The incident category with the highest number of records among the four selected crime types.

7. What This Project Can Be Used For

The analysis can support **urban safety monitoring, patrol visibility planning, risk communication, and resource allocation** by showing when and where selected incidents concentrate. It is a descriptive and exploratory analysis: the code identifies historical patterns in the chosen six-month window; it does not prove causation or guarantee future crime.

8. Important Method Assumption

The code uses 0.0045 degrees as an approximation for 500 meters inside ST_DWithin while the coordinates are in latitude/longitude (EPSG:4326). Therefore, the 500-meter boundary should be interpreted as an approximate spatial filter as implemented in the notebook, rather than an exact geodesic distance calculation.

One-line Project Summary

Analyzed six months of selected Philadelphia theft and burglary incidents within an approximately 500-meter urban zone, using spatial filtering, hourly/day-of-week analysis, interactive mapping, and heatmaps to identify incident concentration, peak time periods, and dominant crime patterns.